Greenness Benchmarking for Web Frameworks and Best Practices

A Procurement Platform for Indigenous Businesses as a Case Study

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1. Background

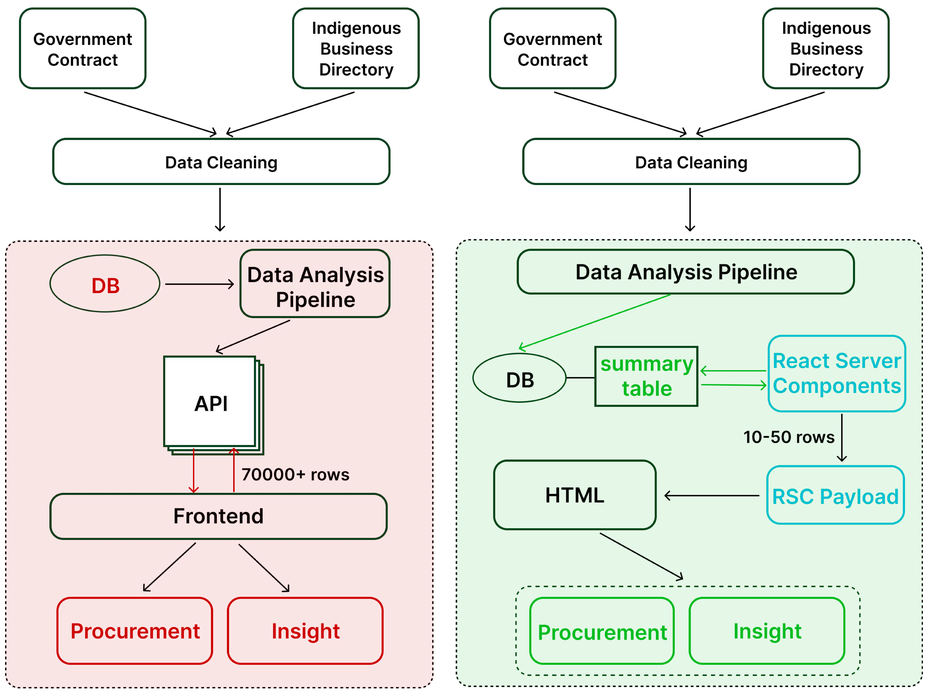
Research Gaps

- evaluation methodologies **without real-world examples**
- focus on listing best practices but **without quantifying the impact**

Research Questions

- measure greenness and performance by **adopting existing tools**
- compare and contrast at both **framework and component level**

2. Architecture Design



3. Evaluation

Lighthouse Report

Procurement without pagination

TBT: Total Blocking Time
FCP: First Contenful Paint
SI: Speed Index
LCP: Largest Contenful Paint

TBT	FCP	SI	LCP
470 ms	0.9 s	0.9 s	1.2 s

Procurement with pagination

TBT	FCP	SI	LCP
0 ms	0.3 s	0.5 s	0.5 s

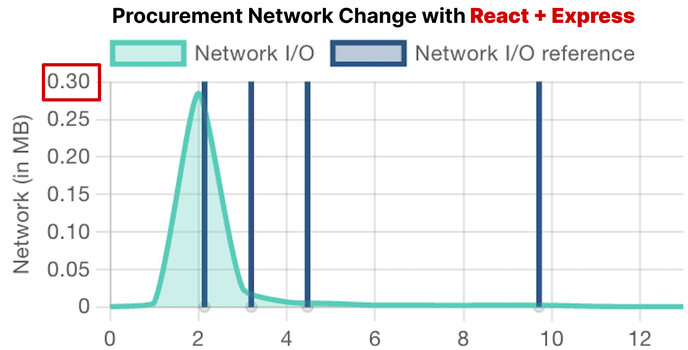
Insight with React + Express

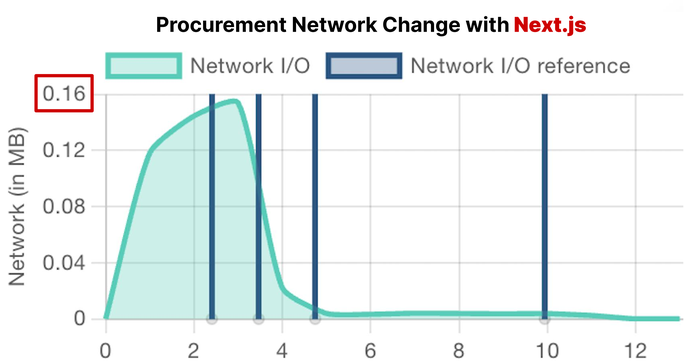
TBT	FCP	SI	LCP
80 ms	0.5 s	1.1 s	1.1 s

Insight with Next.js

TBT	FCP	SI	LCP
0 ms	0.3 s	0.8 s	0.5 s

GreenFrame Analysis





Summary

Procurement	CPU in mWh	Network in mWh	Memory in GB	FCP in s	LCP in s	TBT in ms
React + Express + pagination	3	3.7	0.4	0.5	0.5	0
Next.js + pagination	4.3	5.1	0.41	0.3	0.5	0
Next.js	3.9	18	0.51	0.9	1.2	470

Insight	CPU in mWh	Network in mWh	Memory in GB	FCP in s	LCP in s	TBT in ms
React + Express	2.6	3.5	0.18	0.5	1.1	80
Next.js	2.8	4.3	0.18	0.3	0.5	0

Conclusion

Next.js improves the **responsiveness** of web applications, especially during the **initial load**, but with a **slight increase in energy consumption.**

Server-side pagination is essential to avoid significant **network overhead** when handling **large datasets.**

4. Demo

5. Future Work

Currently only evaluating frontend energy consumption, and the impact made by summary tables is not fully captured.

Test with other similar tools to further validate our findings.